FREELANCE MARKETPLACE

Data Flow Diagram (DFD)

System Process Documentation

Technology Stack: Python + Django + MySQL

Version 1.0 | January 2025

Academic Project Documentation

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1. Introduction

This document presents the Data Flow Diagrams (DFD) for the Freelance Marketplace application. DFDs illustrate how data moves through the system, showing processes, data stores, external entities, and the flow of information between them.

The diagrams are presented in three levels: Context Diagram (Level 0), Level 1 DFD showing major processes, and Level 2 DFDs showing detailed sub-processes for key modules.

2. DFD Notation

The following notation is used in the Data Flow Diagrams:

Symbol	Name	Description
□	External Entity	Sources or destinations of data outside the system
○	Process	Transforms input data to output data
═══	Data Store	Repository of data (database tables)
→	Data Flow	Movement of data between components

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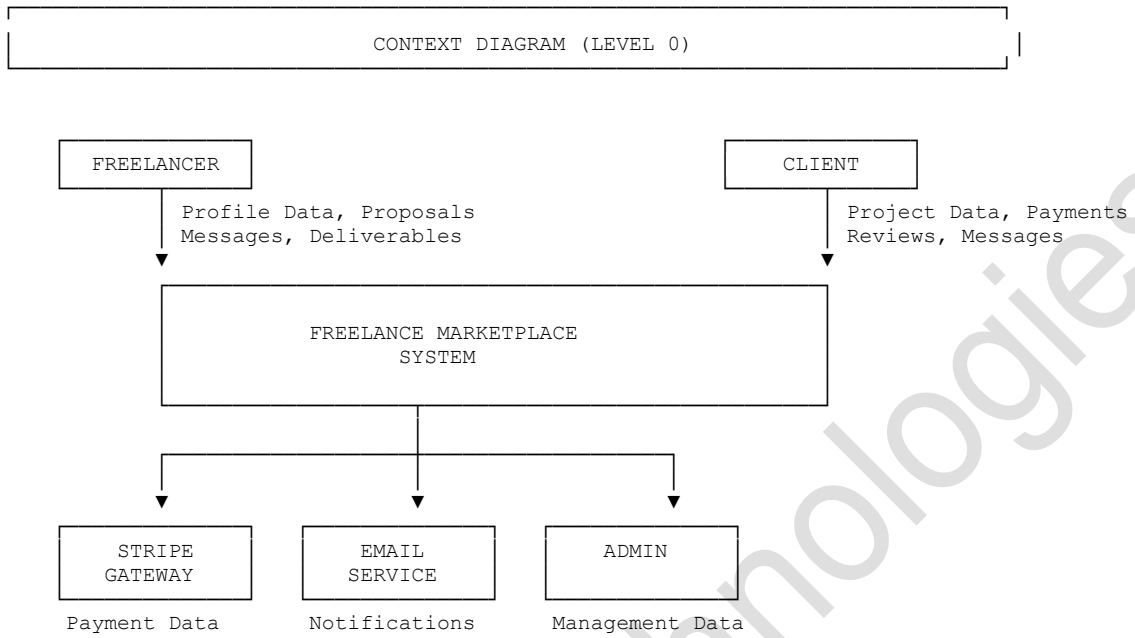
3. Context Diagram (Level 0)
The Context Diagram shows the system as a single process with its external entities:



3.1 External Entities

Entity	Description
Freelancer	Professionals who create profiles, submit proposals, and deliver work
Client	Individuals/businesses who post projects, hire freelancers, and make payments
Admin	Platform administrators who manage users, resolve disputes, and monitor activity
Stripe Gateway	External payment processing service for escrow and transfers
Email Service	External SMTP service for sending notifications and verifications

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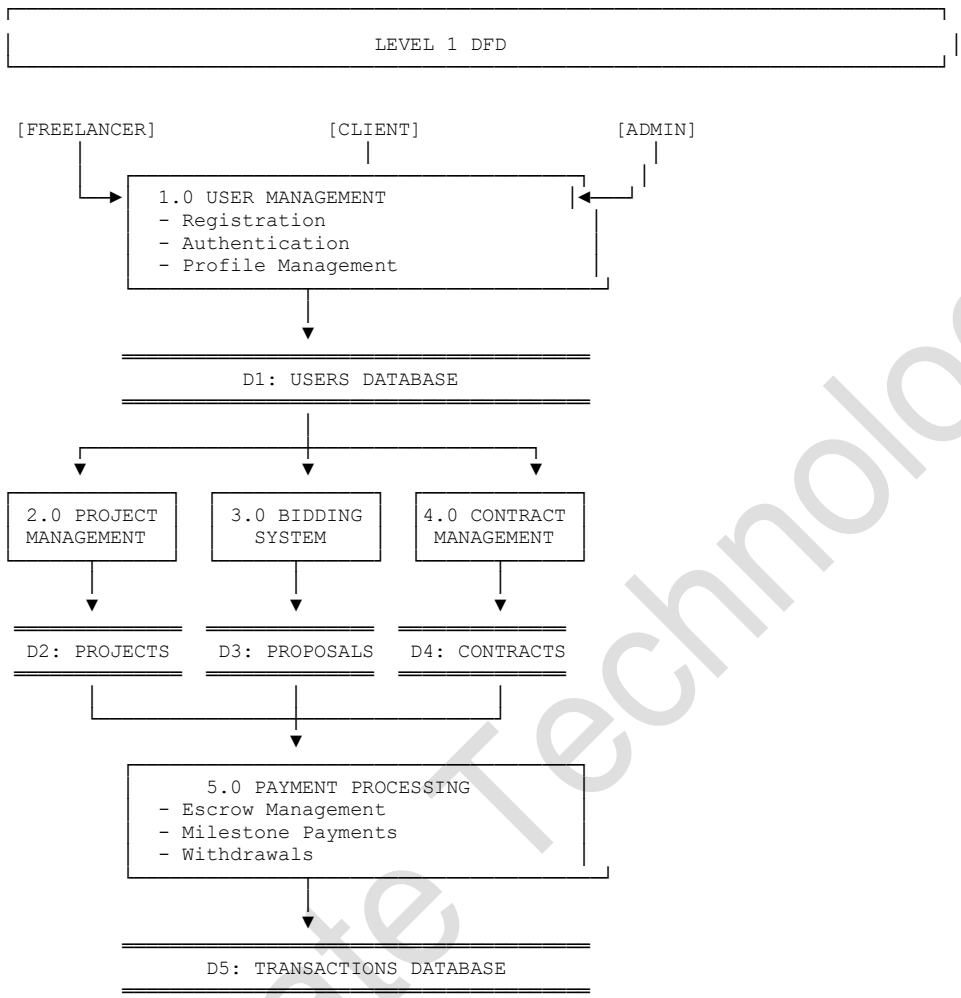
4. Level 1 DFD

The Level 1 DFD breaks down the system into major processes:



4.1 Process Descriptions

ID	Process	Description
1.0	User Management	Handles user registration, authentication, profile creation and updates
2.0	Project Management	Manages project creation, publishing, search, and status updates
3.0	Bidding System	Processes proposal submissions, reviews, and acceptance/rejection
4.0	Contract Management	Manages contracts, milestones, deliverables, and completion

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5.0	Payment Processing	Handles escrow funding, milestone releases, and withdrawals
6.0	Messaging	Manages real-time conversations and message delivery
7.0	Review System	Processes reviews and calculates user ratings
8.0	Notification System	Generates and delivers notifications via multiple channels

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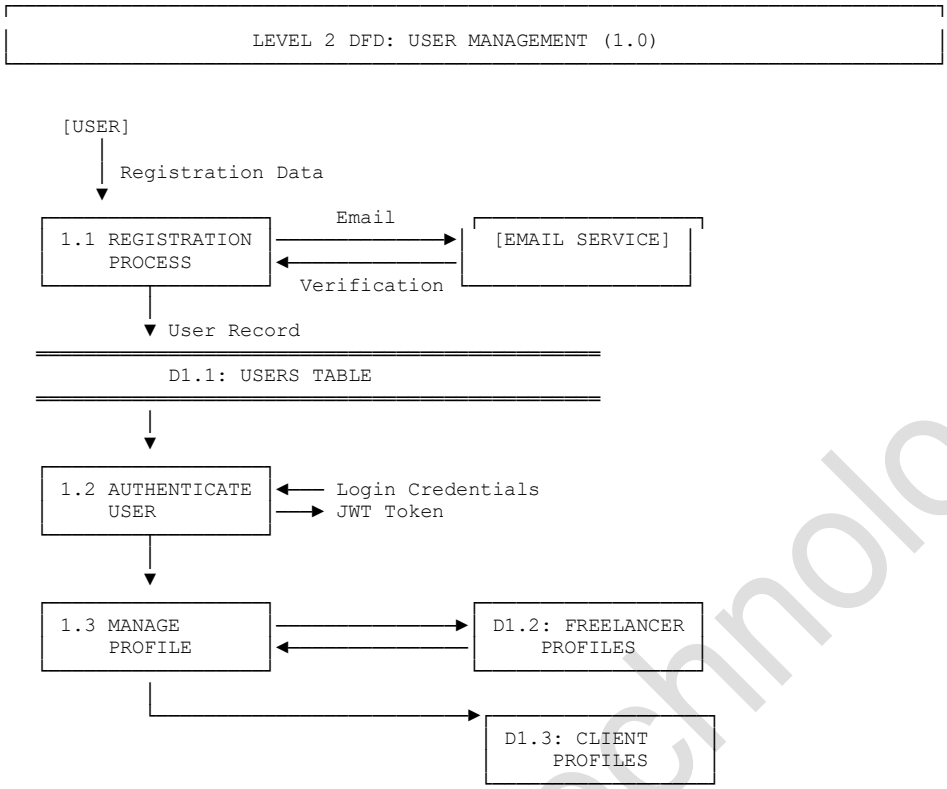
5. Level 2 DFD - User Management



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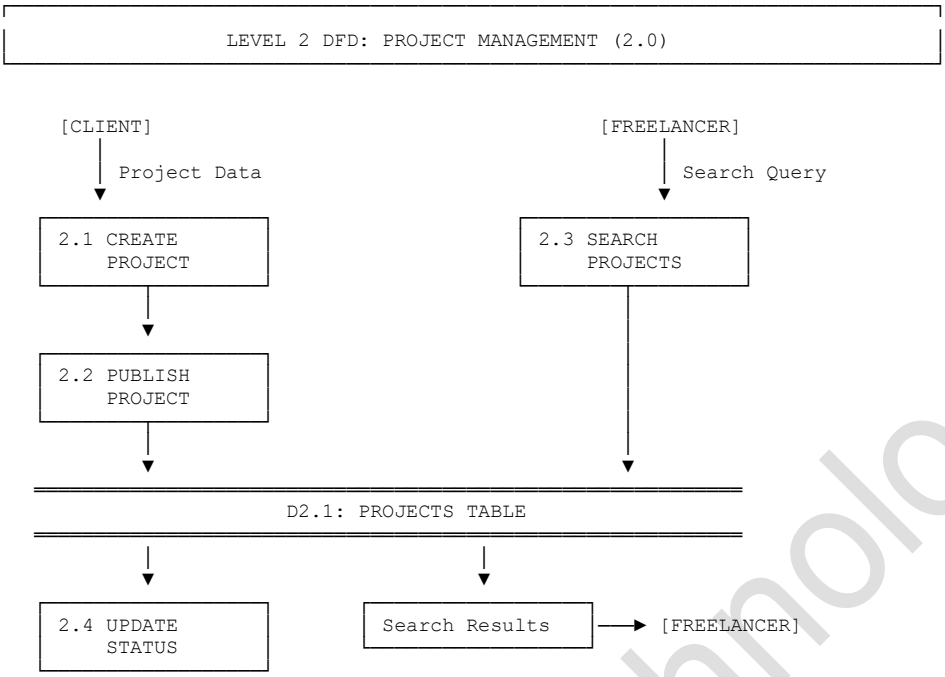
6. Level 2 DFD - Project Management



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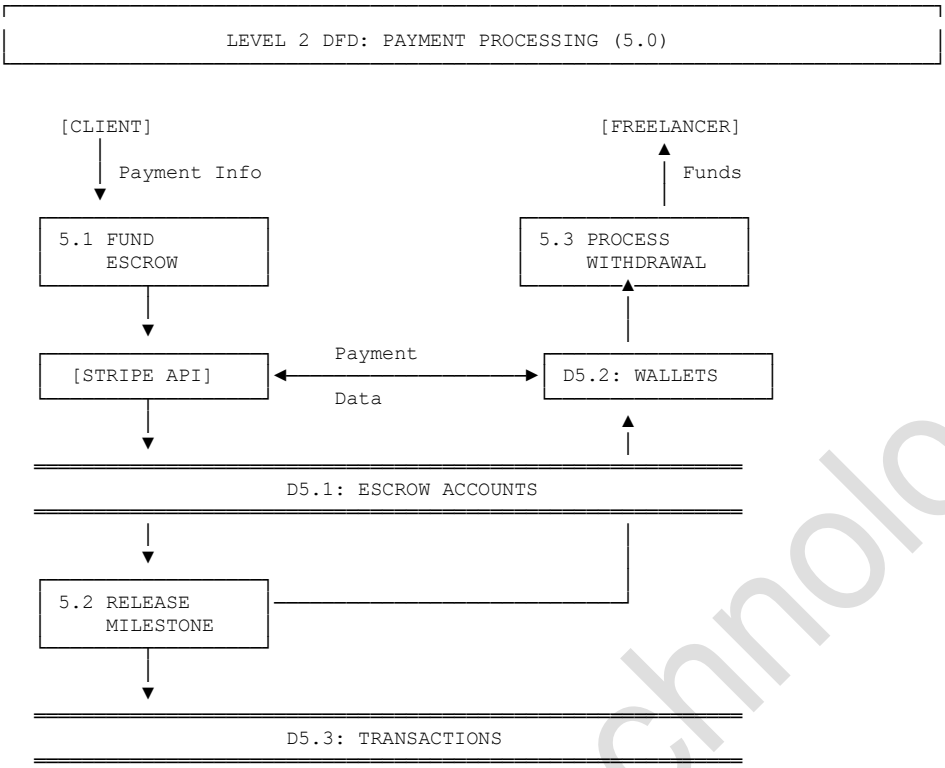
7. Level 2 DFD - Payment Processing



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8. Data Dictionary

8.1 Data Stores

ID	Data Store	Description
D1	Users Database	Stores user accounts, credentials, and authentication data
D1.1	Freelancer Profiles	Extended profile data for freelancers (skills, rates, portfolio)
D1.2	Client Profiles	Extended profile data for clients (company info, spending)
D2	Projects	Project listings with requirements, budget, and status
D3	Proposals	Freelancer bids with cover letters and pricing
D4	Contracts	Active contracts with terms and milestones
D5	Transactions	Payment records, escrow, and withdrawal history
D6	Messages	Conversation threads and individual messages
D7	Reviews	User reviews with ratings and feedback
D8	Notifications	User notifications and delivery status

8.2 Data Flows

Data Flow	Source → Destination	Data Elements
Registration Data	User → 1.1	email, password, name, role
Login Credentials	User → 1.2	email, password
JWT Token	1.2 → User	access_token, refresh_token, expiry
Project Data	Client → 2.1	title, description, budget, skills, deadline
Proposal Data	Freelancer → 3.1	cover_letter, bid_amount, duration
Payment Info	Client → 5.1	amount, payment_method, contract_id
Message Content	User → 6.1	content, attachments, conversation_id
Review Data	User → 7.1	rating, comment, contract_id

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